

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Clarification for the Recovery Flow
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

This ECN emphasizes the fact that the whole enabled transmitters should be turned off before the transition from Recovery Flow (CLd) to Initialization Flow (Training).

Benefits as a result of the changes:

Eliminates potential races and corner cases by adding a clear sync event between moving to Initialization Flow.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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NA

An analysis of the hardware implications:
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NA

An analysis of the software implications:
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NA

An analysis of the compliance testing implications:
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NA

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Actual Change

(a). Section 4.4.7 Gen 4 Link Recovery

To Text:

After a Router receives the last bit of an ELT_Recovery Transaction, it shall:

- Turn off all enabled transmitters within tRecoveryTxOff. The Lane Common Mode voltages do not need to be maintained.
- Send an ELT_Recovery Sideband Transaction within tRecoveryResponse. This step is skipped if the Router already sent an ELT_Recovery Transaction as part of the Link Recovery initiation sequence.

After both sending and receiving the last bits of the ELT_Recovery Transactions and all enabled transmitters are turned off, a Router shall:

- Start Lane Initialization from Phase 4 with the following changes:
 - The Lane Adapters shall use the last set of TxFFE parameters used prior to Gen 4 Link Recovery without performing the TxFFE negotiation flow.
Note: It is not required to write the Start TxFFE bit as mentioned in Section 4.1.2.4 step 4.a.
 - It is recommended to minimize the time interval between Broadcast RT Sideband Transactions in order to avoid expiration of the tRecoveryAbort timer.